SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I **__Bandaru Jayasri-192424301__** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
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Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → ______ → Transport → Network → ______ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

Given Concept Map

Application
↓
Presentation
↓
Session
↓
Transport
↓
Network
↓
Data Link
↓
Physical

Missing Layers

- **Session Layer**
- **Data Link Layer**

Explanation

Session Layer

- Establishes, manages, and terminates communication sessions. Provides synchronization and recovery during data transfer.

Data Link Layer

- Converts packets into frames.
- Performs framing, error detection, and node-to-node delivery using MAC addresses.

Importance for Reliable Communication

- The Session Layer ensures uninterrupted communication through synchronization.
- The Data Link Layer detects transmission errors and delivers frames reliably between adjacent devices.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

Application
Presentation
Session
|
▼
Transport
|
Segment
|
▼
Network
|
Packet
|
▼
Data Link
|
Frame
|
▼
Physical
|
Bits

Analysis

OSI Layer	Data Unit	Function
Transport	Segment	Divides data into segments
Network	Packet	Adds IP address and routes data
Data Link	Frame	Adds MAC address and error detection
Physical	Bits	Converts data into electrical/optical/radio signals

Conclusion

As data moves downward through the OSI layers, headers are added (encapsulation). At the receiver, headers are removed (decapsulation), ensuring successful communication.

3.Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

Concept Map

Data Link Layer
|
▼
MAC Address
|
▼
Local Delivery

Network Layer
|
▼
IP Address
|
▼
Global Delivery

Explanation

- **MAC Address**
  - Used for communication within the same local network (LAN).
  - Ensures correct device-to-device delivery.
- **IP Address** ○ Used for communication across different networks.
  ○ Enables routers to forward packets to the correct destination.

Analysis

The Network Layer delivers packets to the correct network using IP addresses, while the Data Link Layer delivers frames to the correct device using MAC addresses. Together, they ensure complete end-to-end communication.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

```
Error Control
|
┌────────────┴────────────┐
|            |
▼            ▼
Data Link Layer     Transport Layer
|            |
Error Detection     Error Correction
|            |
└──────────Reliable Communication──────────┘
```

Analysis

Data Link Layer

- Detects transmission errors using CRC or checksum.
- Requests retransmission between directly connected devices.

Transport Layer

- Performs end-to-end error recovery.
- Uses acknowledgements (ACKs), sequence numbers, and retransmissions.

Conclusion

The Data Link Layer detects errors locally, while the Transport Layer ensures complete errorfree delivery between source and destination.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

User

│
▼
Application
(HTTP/HTTPS)
│
▼
Presentation
(Encryption - SSL/TLS)
│
▼
Session
(Session Management)
│
▼
Transport
(TCP)
│
▼
Network
(IP Routing)
│
▼
Data Link
(Frame Transmission)
│
▼
Physical (Cables/Wi-Fi)

Analysis

Layer Failure	Effect
Physical	No signal transmission
Data Link	Frame errors and communication failure
Network	Packets cannot reach destination
Transport	Data loss and connection failure
Session	Session disconnected
Presentation	Encryption/decryption fails
Application	Web page cannot load

Conclusion

All OSI layers work together. Failure in one layer interrupts the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✖ → Transport ✖ → Application ✖

Analyze the problem and explain how the concept map helps in troubleshooting.

Given Concept Map

```
Physical     ✓
    |
Data Link    ✓
    |
Network      ✕
    |
Transport    ✕
    |
Application  ✕
```

Analysis

- Physical Layer is working correctly.
- Data Link Layer is functioning properly.
- Failure begins at the **Network Layer**.

Possible Causes

- Incorrect IP address
- Router failure
- Incorrect subnet mask
- Gateway configuration error
- Routing problem

Since the Network Layer fails:

- Transport Layer cannot establish TCP/UDP communication.
- Application Layer cannot access network services.

Troubleshooting Steps

1. Check IP address configuration.
2. Verify default gateway.
3. Test connectivity using **ping**.
4. Verify router status.
5. Check routing tables.
6. Trace the path using **traceroute**.

Conclusion

The concept map helps identify that the problem starts at the **Network Layer**, making troubleshooting faster and more systematic.

RUBRICS FOR CALCULATION

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation